

Fig. 2: Power flow in a typical electric vehicle with supercapacitor

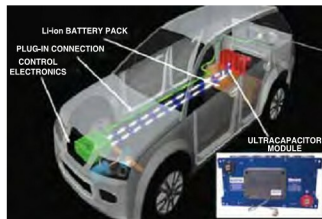


Fig. 3: Concept car using Li-ion battery pack with ultracapacitor module

Li-ion Battery vs Electrochemical or Supercapacitor

Characteristic	State-of-the-art Lithium-Ion battery	Electrochemical capacitor
Charge time	3-5 minutes	1 second
Discharge time	3-5 minutes	1 second
Cycle life	<5000 @ 1C rate	>500,000
Specific energy (Wh/kg)	50-100	5
Specific power (kW/kg)	1-2	5-10
Cycle efficiency (per cent)	<50% to >90%	<75% to >95%
Cost/Wh	\$0.5-1/Wh	\$10-20/Wh
Cost/kW	\$50-150/kW	\$15-30/kW

storage devices are shown in Regon's plot of power density versus energy density.

Fig. 2 shows the block diagram of

2. Time constant (ESR and capacitance value)
3. Output voltage and output current

power flow in typical electrical vehicles. The combination of a primary energy source and an ultracapacitor takes care of the average power and peak power demand, respectively.

Fig. 3 shows the concept car having the Li-ion battery pack with ultracapacitor. The Li-ion battery will replace the conventional lead-acid battery in the near future. Considering various parameters of the supercapacitor, it will be the next-generation EES device.

Design parameters

1. Trade-off between energy and power

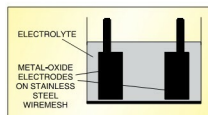


Fig. 4: Construction of a supercapacitor

4. Use of different electrolyte and self-discharge

Application areas

1. Quick-charge applications – charging in seconds and discharging in minutes (required for power tools and toys)
2. Electric vehicles (starting/regenerative braking)
3. At sharp turns on electric traction
4. Short-term support for uninterrupted power system with generator backup
5. Solar array (energy-rich, poor power source)
6. Supporting power buses while switching
7. Computer memory backups – low-current, long-duration requirements

Summing up

Energy storage is a persistent problem in electrical systems. There are many energy storage systems, but electrical energy can be stored only in two fundamentally different ways: battery and capacitor. Scientists and engineers are looking for devices beyond these.

It has been found that a supercapacitor along with conventional battery can be a better electrical energy storage device than the battery alone. Though the supercapacitor technology is at present in infancy, it stands a bright chance of becoming the next-generation electrical energy storage device. ●

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